



Existence and Uniqueness for Electrode Models for Electric Current Computed Tomography

Author(s): Erkki Somersalo, Margaret Cheney and David Isaacson

Source: *SIAM Journal on Applied Mathematics*, Aug., 1992, Vol. 52, No. 4 (Aug., 1992), pp. 1023-1040

Published by: Society for Industrial and Applied Mathematics

Stable URL: <https://www.jstor.org/stable/2102189>

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at <https://about.jstor.org/terms>



JSTOR

Society for Industrial and Applied Mathematics is collaborating with JSTOR to digitize, preserve and extend access to *SIAM Journal on Applied Mathematics*

EXISTENCE AND UNIQUENESS FOR ELECTRODE MODELS FOR ELECTRIC CURRENT COMPUTED TOMOGRAPHY*

ERKKI SOMERSALO†, MARGARET CHENEY‡§, AND DAVID ISAACSON‡¶

Abstract. The following experiment is considered. To a body of given conductivity, a number of electrodes are attached, through which current is sent. On the same electrodes, the resulting voltages are measured. This experiment can be described by a number of mathematical models [K.-S. Cheng et al., *IEEE Transactions on Biomedical Engineering*, 36 (1989), pp. 918–924]. These models are discussed and their predictions compared with experiment. In particular, a model is exhibited that is capable of predicting the experimentally measured voltages to within 0.1 percent. For this model, existence and uniqueness of the associated electrical potential is proved.

Key words. impedance imaging, electrical impedance tomography, electrical current computed tomography, electrode models, variational principles

AMS(MOS) subject classifications. 35J25, 78A70

1. Introduction. We consider the following experiment. On the surface $\partial\Omega$ of a body Ω of conductivity σ , we attach L electrodes. Through these electrodes, we send current into the body, and, on these same electrodes, we measure the resulting voltages.

In this paper we discuss a number of mathematical models for this experiment [1]. In § 2 we discuss each model and compare its predictions with experimental results. In particular, we exhibit a model that is capable of predicting the measured voltages to at least 0.1 percent. In § 3 we prove the existence and uniqueness of solutions to this model in a specific class of functions. In § 4 we discuss a symmetry property of the model. The paper concludes with three appendices. The first two give technical details, and the third shows how the ideas of § 3 can be used for numerical computation in a simple case.

We identify the electrode with the part of the surface it contacts. These subsets of $\partial\Omega$ we denote by e_l , $1 \leq l \leq L$. We denote the current applied to e_l by I_l . Let $I = (I_1, \dots, I_L)^T$, where the superscript T denotes transpose. We call a vector of L currents a “current pattern” if it satisfies the conservation of charge condition $\sum_{l=1}^L I_l = 0$. The corresponding “voltage pattern” we denote by $V = (V_1, \dots, V_L)^T$. We choose the ground or reference potential, so that $\sum_{l=1}^L V_l = 0$. The relationship between I and V is experimentally determined to be a linear one. Thus we can write

$$(1.1) \quad V = RI,$$

where R is an $L \times L$ matrix called the resistance matrix. It satisfies the following symmetry relation:

$$(1.2) \quad R = R^T.$$

* Received by the editors January 28, 1991; accepted for publication (in revised form) October 15, 1991.

† Department of Mathematics, University of Helsinki, Hallituskatu 15, 00100 Helsinki, Finland. This work was done while this author was visiting Rensselaer Polytechnic Institute, Troy, New York 12180.

‡ Department of Mathematical Sciences, Rensselaer Polytechnic Institute, Troy, New York 12180.

§ This author’s research was supported in part by Office of Naval Research grant N-00014-89-J-1129 and by funds from Duke University under the Office of Naval Research Young Investigator grant N00014-85-K-0224.

¶ This author’s research was supported in part by National Institutes of Health grants GM 39388 and GM 42935, National Science Foundation grant BCS-8706340, General Electric, and the Rensselaer Polytechnic Institute.

TABLE 1
The characteristic resistances in the column labelled ρ_k^{exp} were measured experimentally. The numbers in the column labelled ρ_k were computed from the complete model. This work is described in more detail in [1].

$\rho_0(\Omega\text{-cm})$	284.0		139.7		62.3		29.5	
$z(\Omega\text{-cm}^2)$	58.0		30.5		15.0		7.5	
k	ρ_k^{exp}	ρ_k	ρ_k^{exp}	ρ_k	ρ_k^{exp}	ρ_k	ρ_k^{exp}	ρ_k
1	312.0	312.0	156.8	156.8	70.7	70.7	33.9	33.9
2	150.0	148.4	75.0	74.4	33.8	33.7	16.3	16.1
3	96.0	94.8	47.9	47.5	21.5	21.5	10.1	10.3
4	69.5	68.7	34.6	34.4	15.7	15.6	7.6	7.5
5	54.0	53.5	27.0	26.8	12.2	12.2	5.9	5.8
6	44.1	43.7	22.0	21.9	10.0	10.0	4.8	4.8
7	37.3	37.0	18.6	18.6	8.5	8.5	4.1	4.1
8	32.4	32.3	16.2	16.2	7.4	7.4	3.6	3.6
9	28.9	28.8	14.4	14.4	6.6	6.6	3.2	3.2
10	26.2	26.2	13.1	13.1	6.0	6.0	2.9	2.9
11	24.3	24.2	12.2	12.1	5.6	5.6	2.7	2.7
12	22.8	22.8	11.4	11.4	5.2	5.2	2.6	2.5
13	21.8	21.7	10.9	10.9	5.0	5.0	2.5	2.4
14	21.0	21.0	10.6	10.6	4.9	4.9	2.4	2.3
15	20.6	20.6	10.3	10.4	4.8	4.8	2.3	2.3
16	20.5	20.5	10.3	10.3	4.7	4.7	2.3	2.3

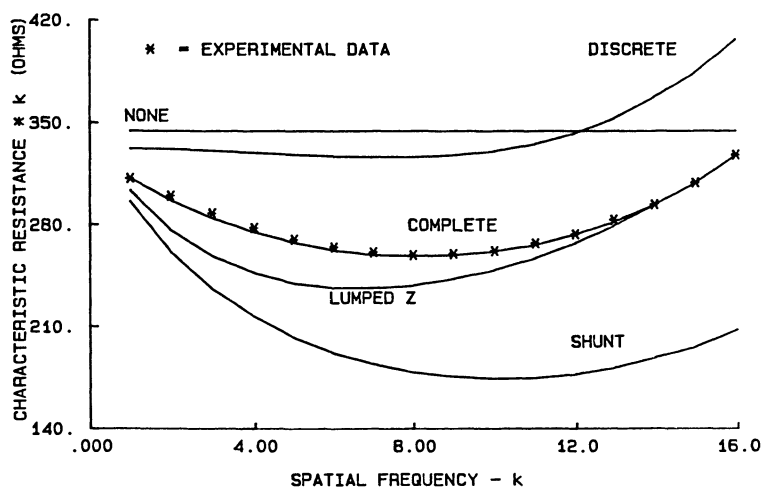


FIG. 1. The product of characteristic resistance times spatial frequency is plotted versus spatial frequency for spatial frequencies between 1 and 16. Experimental measurements made with a homogeneous saline phantom of 284 $\Omega\text{-cm}$ resistivity are shown as asterisks. The horizontal line labelled “NONE” is the model with no electrodes considered. The curve labelled “DISCRETE” corresponds to the gap model, which includes the effect of discretization of the current pattern. The curve “SHUNT” models perfectly conducting electrodes that shunt current across their surface. The curved “LUMPED Z” has a lumped electrode resistance added to each of the perfectly conducting electrodes. The curve through the asterisks, labelled “COMPLETE,” includes the effects of discretization, shunt, and a surface impedance of 58 $\Omega\text{-cm}^2$.

The nonzero eigenvalues of R are called the characteristic resistances, and the corresponding eigenvectors are the characteristic currents. Since the $L-1$ characteristic currents form a complete set of current patterns, knowing them and the corresponding characteristic resistances enables us to predict the voltages that result from applying any current pattern.

In some special cases, the characteristic current patterns can be found explicitly. One such case occurs when the body in question is a cylinder whose conductivity in the interior is constant, around the boundary of which have been placed equally spaced, identical electrodes. In this case, the experiment is invariant under discrete rotations in the following sense. If we apply a particular current pattern, and then apply the same pattern rotated by, say, one electrode, the resulting voltage patterns differ only in that the second is a rotated version of the first. Mathematically, this can be expressed by saying that the matrix R commutes with discrete rotations. Under these circumstances, the characteristic currents can be computed explicitly: when L is even, they are

$$(1.3) \quad T_l^k = \begin{cases} \cos k\theta_l, & 1 \leq l \leq L, \quad 1 \leq k \leq \frac{L}{2}, \\ \sin\left(\frac{L}{2} - k\right)\theta_l, & 1 \leq l \leq L, \quad \frac{L}{2} < k \leq L-1, \end{cases}$$

where $\theta_l = 2\pi l/L$. (These computations are carried out in Appendix 1.)

Once these characteristic current patterns are known, an arbitrary current pattern I can be expanded as

$$I = \sum_{k=1}^{L-1} \langle I, T^k \rangle^k,$$

where $\langle \cdot, \cdot \rangle$ denotes the inner product in R^L . The voltages due to I are then

$$RI = \sum_{n=1}^{L-1} \langle I, T^k \rangle RT^k.$$

Since $RT^k = \rho_k T^k$, we have reduced the prediction of the outcome of every experiment to the knowledge of the characteristic resistances ρ_k , $k = 1, \dots, L-1$. Since $L-2$ of the characteristic resistances are doubly degenerate, we need only compute $(L-2)/2 + 1$ characteristic resistances to predict the outcome of every experiment.

The characteristic resistances were measured experimentally by Cheng et al. [1]. Thirty-two metal electrodes were placed, equally spaced, around the circumference of a cylindrical tank of radius $r_0 = 15$ cm. Four different experiments were performed; in each, the tank was filled to depth of 3 cm with salt water of different concentrations. The different concentrations result in different resistivities $1/\sigma_0$, which, for these experiments, were between 29.5 and 284 ohm-cm. (See Table 1.)

To determine the voltage due to any current pattern, i.e., the operator R , Cheng et al. applied the independent patterns of current and measured the resulting $L-1$ voltage patterns. From these measurements, the $L/2$ characteristic resistances were found.

Figure 1 shows the computed characteristic resistances for the different mathematical models compared with experimental data. In the next section, we explain these models.

2. The models. It is relatively straightforward to model the electromagnetic fields within the body: we use Maxwell's equations, which are

$$(2.1) \quad \nabla \wedge E = -\partial_t B,$$

$$(2.2) \quad \nabla \wedge H = J + \partial_t D,$$

$$(2.3) \quad \nabla \cdot B = 0,$$

$$(2.4) \quad \nabla \cdot D = \rho,$$

where, for a linear, isotropic, nondispersive body,

$$(2.5) \quad D = \varepsilon E,$$

$$(2.6) \quad B = \mu H,$$

$$(2.7) \quad J = \sigma E.$$

Here E is the electric field, B the magnetic field, ρ the charge density, J the current, ε the electric permittivity, μ the magnetic permeability, and σ the conductivity. We note that conductivity is the reciprocal of resistivity.

There are two possible approaches to obtaining a simpler differential equation for the interior of the body. One approach is to assume that our experiments are static ones. This allows us to set the time derivatives in (2.1) and (2.2) to zero. From (2.1) we then conclude that there is a potential u , called the electric potential, such that

$$(2.8) \quad E = -\nabla u.$$

Taking the divergence of (2.2) and using (2.7) and (2.8), we conclude that

$$(2.9) \quad \nabla \cdot \sigma \nabla u = 0.$$

This is the differential equation that we use in the interior of the body.

The second approach to obtaining an equation like (2.9) is to look for time-harmonic solutions of (2.1)–(2.7) and, at the same time, assume that the magnetic permeability μ is very small. Thus we can expand (2.1)–(2.7) in power series in μ (this is done in Appendix 2); one of the resulting zero-order equations is (2.9) with σ replaced by $\sigma + i\omega\varepsilon$, where ω is the frequency. To simplify notation, in the remainder of the paper, we simply consider (2.9), where σ is allowed to be complex.

Boundary conditions, on the other hand, are more complicated. If we knew the current density j on the boundary, then an appropriate boundary condition would be

$$(2.10) \quad \sigma \frac{\partial u}{\partial \nu} = j,$$

where ν is the outward unit normal. How should we choose j ? A naive choice would be to assume that j is a continuous function. For the experiments on the homogeneous tank, (1.3) suggests that j could be $\sin k\theta$ or $\cos k\theta$. In this case, (2.9) and (2.10) form what we call the continuum model, or no-gap model. It has the advantage of being easy to work with mathematically. For example, for a homogeneous tank with radius r_0 , we can use separation of variables to calculate [2] that the characteristic resistances are $\rho_k = r_0(\sigma_0 k)^{-1}$, $k = 1, 2, \dots, L/2$. These characteristic resistances, multiplied by k , appear in Fig. 1 as the curve labelled “none.” This curve does not reproduce the experimental data. It overestimates the resistivity by as much as 25 percent.

A better model is the so-called gap model, in which j is assumed to be

$$(2.11) \quad j = \begin{cases} \frac{I_l}{|e_l|} \text{ on } e_l, & l = 1, 2, \dots, L, \\ 0 & \text{off } \bigcup_{l=1}^L e_l, \end{cases}$$

where $|e_l|$ is the area of the electrode. For the homogeneous tank experiments, I_l is given by (1.3). Again, the model has the advantage of being easy to work with

mathematically. For the homogeneous tank, we can still use separation of variables to calculate the characteristic resistances [2]. The results of these calculations, multiplied by k , appear in Fig. 1 as the curve labelled “discrete.” We see that the gap model gives only a slight improvement over the continuum model. Both the continuum model and the gap model overestimate the resistivity of the tank because they ignore the shorting or shunting effect of the electrodes. That is, the metal electrodes themselves provide a low-resistance path for the current.

This shorting effect is accounted for in the so-called shunt model, which makes the standard assumption that the metal is a perfect conductor, and hence the potential on the metal is constant. Once we constrain the boundary values of the potential, however, attempts to specify the current density by (2.10) result in an overdetermined problem. We therefore replace (2.10) by the following weaker conditions, which are all that we really know about the current density:

$$(2.12) \quad \int_{e_l} \sigma \frac{\partial u}{\partial \nu} dS = I_l, \quad l = 1, 2, \dots, L$$

and

$$(2.13) \quad \sigma \frac{\partial u}{\partial \nu} = 0 \quad \text{off } \bigcup_{l=1}^L e_l.$$

We write the condition that the potential be constant on each electrode as

$$(2.14) \quad u = U_l \quad \text{on } e_l, \quad l = 1, 2, \dots, L,$$

where the constants U_l are not specified in advance; they are the measured voltages and are part of the solution of the problem.

The shunt model is thus composed of (2.9), (2.12), and (2.13), together with the constraints (2.14). It is widely used in modelling electrodes; see [6] for a recent example. This model has the disadvantage of no longer being a standard Neumann problem. It is, however, properly formulated if we include conservation of charge

$$(2.15) \quad \sum_{l=1}^L I_l = 0,$$

which is needed for existence of a solution, and a condition on the voltages to specify the ground or zero potential, which is needed for uniqueness of the solution. The simplest of these conditions is

$$(2.16) \quad \sum_{l=1}^L U_l = 0.$$

The shunt model has the added disadvantage of not reproducing the experimental tank data to the precision that we can measure. This can be seen from the curve labelled “shunt” in Fig. 1. We note that the characteristic resistances predicted by the shunt model are less than the experimental ones. Also, the discrepancy gets worse at higher spatial frequencies. Because the higher spatial frequencies contain proportionately more information about the outside edge of the tank, we suspect that there is an extra, unaccounted-for resistance between the electrode and saline. This resistance is due to an electrochemical effect at the saline-electrode interface. In particular, at this interface there forms a thin, highly resistive layer that we characterize by a quantity z_l . We call z_l the effective contact impedance or “surface impedance.” The voltage drop across

this layer is the product of the surface impedance z_l times the current density $\sigma(\partial u/\partial \nu)$. Thus (2.14) should be replaced by

$$(2.17) \quad u + z_l \sigma \frac{\partial u}{\partial \nu} = U_l \quad \text{on } e_l, \quad l = 1, 2, \dots, L.$$

This gives us what we call the complete model or the shunt-plus-surface-impedance model. Its predictions for the homogeneous rank are shown by the curve labelled “complete” in Fig. 1. We see that this model predicts the voltages to the precision with which we can measure. In the next section, we prove existence and uniqueness of solutions to the complete model.

The variational formulation that we use to prove existence and uniqueness also forms the basis for stable numerical methods for the calculation of the voltages U_l . A simple calculation illustrating such a method is given in Appendix 3.

In particular, Appendix 3 shows how to calculate the characteristic resistances ρ_l for the complete model in the simple case needed for the experiments summarized in Fig. 1. In these experiments, the conductivity is constant, and the tank is invariant under discrete rotations. The surface impedances z_l are therefore all identical. This single value z is determined by computing ρ_{16} for a succession of values of z and choosing z so that the computed value agrees with the experimentally determined one. We choose to fit ρ_{16} , the characteristic resistance of highest spatial frequency, because it is the one least sensitive to changes in the interior. The graph shows that once this single surface impedance is determined, the other 15 characteristic resistances agree with the experimentally measured ones to 0.1 percent, which is the measurement precision of the instrument.

3. Existence and uniqueness for the complete model. We start this section by defining some notation and specifying the assumptions about the domain Ω and the (complex) conductivity σ . The general reference in this introductory part is [4].

Let Ω be a bounded domain in $\mathbb{R}^n$, $n = 2, 3$, with a smooth boundary $\partial\Omega$. We denote a point in Ω by x . The electrodes e_l , $1 \leq l \leq L$ are assumed to be open connected subsets of $\partial\Omega$ whose closures are disjoint. In the case where $n = 3$, we assume for simplicity that the boundaries of the electrodes are smooth curves on $\partial\Omega$.

We also assume that the conductivity σ is continuously differentiable in Ω up to the boundary; i.e., the boundary $\partial\Omega$ has a neighborhood U such that $\sigma \in C^1(\bar{\Omega} \cap U)$.

By $H^s(\Omega)$ and $H^s(\partial\Omega)$, $s \in \mathbb{R}$, we mean the L^2 -based Sobolev spaces with smoothness index s over Ω and $\partial\Omega$, respectively. The boundary conditions for the complete model involve restricting u and $\sigma(\partial u/\partial \nu)$ to $\partial\Omega$. The trace mapping γ_0 , defined by

$$(3.1) \quad \gamma_0: u \rightarrow u|_{\partial\Omega}, \quad u \in C^\infty(\bar{\Omega}),$$

extends by a density argument [4] to a continuous mapping

$$(3.2) \quad \gamma_0: H^1(\Omega) \rightarrow H^{1/2}(\partial\Omega).$$

The map $\gamma_1: u \rightarrow \sigma(\partial u/\partial \nu)$ can be defined with the help of Green’s formula. The map is

$$(3.3) \quad \int_{\Omega} v \nabla \cdot \sigma \nabla u \, dx = \int_{\partial\Omega} \sigma \frac{\partial u}{\partial \nu} v \, dS - \int_{\Omega} \sigma \nabla u \cdot \nabla v \, dx,$$

which can be extended to hold for all $u, v \in H^1(\Omega)$, with $\nabla \cdot \sigma \nabla u \in L^2(\Omega)$. For details, we refer to [4].

After these preliminaries, we proceed to the detailed discussion of the complete electrode model. First, we introduce the space

$$(3.4) \quad H = H^1(\Omega) \oplus \mathbb{C}^L.$$

This is the space that contains our solution, which, we recall, involves L surface voltages, as well as the electric potential in the interior. The following proposition states the appropriate weak formulation of the complete electrode model.

PROPOSITION 3.1. *Let Ω , σ and the electrodes e_l satisfy the regularity conditions above, and let $z_l \in \mathbb{C}$, $1 \leq l \leq L$, $\operatorname{Re} z_l > 0$. Assume that $(u, U) = (u(x), (U_l)_{l=1}^L) \in H$, and that u is a (weak) solution to*

$$(3.5) \quad \nabla \cdot \sigma \nabla u = 0 \quad \text{in } \Omega$$

subject to the following boundary conditions on $\partial\Omega$;

$$(3.6) \quad u + z_l \sigma \frac{\partial u}{\partial \nu} = U_l \quad \text{on } e_l, \quad 1 \leq l \leq L,$$

$$(3.7) \quad \sigma \frac{\partial u}{\partial \nu} = 0 \quad \text{on } \partial\Omega \setminus \bigcup_{l=1}^L e_l.$$

Furthermore, we assume that, for a prescribed current pattern $I = (I_l)_{l=1}^L \in \mathbb{R}^L$,

$$(3.8) \quad \int_{e_l} \sigma \frac{\partial u}{\partial \nu} dS = I_l.$$

Then, for any $(v, V) \in H$,

$$(3.9) \quad B((u, U), (v, V)) = \sum_{l=1}^L I_l \bar{V}_l,$$

where $B: H \times H \rightarrow \mathbb{C}$ is the sesquilinear form defined as

$$(3.10) \quad B((u, U), (v, V)) = \int_{\Omega} \sigma \nabla u \cdot \nabla \bar{v} dx + \sum_{l=1}^L \frac{1}{z_l} \int_{e_l} (u - U_l)(\bar{v} - \bar{V}_l) dS.$$

Conversely, if $(u, U) \in H$ satisfies (3.9) for all $(v, V) \in H$, then (u, U) also satisfies (3.5)–(3.8).

Remark. Equations (3.6) and (3.7) can be combined as

$$(3.11) \quad \sigma \frac{\partial u}{\partial \nu} = \sum_{l=1}^L \frac{1}{z_l} (U_l - u) \chi_l,$$

where χ_l is the characteristic function of the l th electrode. This equality holds a priori in $H^{-1/2}(\partial\Omega)$. However, since $u \in H^1(\Omega)$, we have that $U_l - u \in H^{1/2}(\partial\Omega)$ on the boundary, and, by the properties of the Sobolev spaces, (3.11) implies that $\sigma(\partial u / \partial \nu) \in H^s(\partial\Omega)$, where $-\frac{1}{2} \leq s < \frac{1}{2}$. In particular, the integral (3.8) can be interpreted in the classical sense.

Proof of Proposition 3.1. Let $(u, U) \in H$ satisfy (3.5)–(3.8) and let $(v, V) \in H$ be arbitrary. Then we may apply Green's formula (3.3) to obtain

$$(3.12) \quad \int_{\partial\Omega} \sigma \frac{\partial u}{\partial \nu} \bar{v} dS - \int_{\Omega} \sigma \nabla u \cdot \nabla \bar{v} dx = 0,$$

or, from (3.11),

$$(3.13) \quad \sum_{l=1}^L \frac{1}{z_l} \int_{e_l} (U_l - u) \bar{v} dS - \int_{\Omega} \sigma \nabla u \cdot \nabla \bar{v} dx = 0.$$

On the other hand, (3.6) and (3.8) together imply that

$$(3.14) \quad \int_{e_l} u \, dS = \int_{e_l} \left(U_l - z_l \sigma \frac{\partial u}{\partial \nu} \right) dS = U_l |e_l| - z_l I_l,$$

and hence

$$(3.15) \quad \sum_{l=1}^L \frac{1}{z_l} \bar{V}_l \left(\int_{e_l} u \, dS - U_l |e_l| + z_l I_l \right) = 0.$$

Adding (3.15) to (3.13), we get (3.9).

To prove the converse part of the proposition, we assume that $(u, U) \in H$ satisfies (3.9) for all $(v, V) \in H$. Equations (3.5)–(3.8) are then obtained by considering particular choices of (v, V) .

We first choose $v \in C^\infty$ with $\text{supp } v \subset \Omega$ and $V = 0$. Then (3.9) is

$$(3.16) \quad \int_{\Omega} \sigma \nabla u \cdot \nabla \bar{v} \, dx = 0;$$

i.e., in the weak sense, u satisfies $\nabla \cdot \sigma \nabla u = 0$. Therefore by (3.3) we have that

$$(3.17) \quad \int_{\Omega} \sigma \nabla u \cdot \nabla \bar{v} \, dx = \int_{\partial\Omega} \sigma \frac{\partial u}{\partial \nu} \bar{v} \, dS,$$

$v \in H^1(\Omega)$ arbitrary. Combining this with (3.9), where $V = 0$, we have that

$$(3.18) \quad \int_{\partial\Omega} \left(\sigma \frac{\partial u}{\partial \nu} + \sum_{l=1}^L \frac{1}{z_l} \chi_l (u - U_l) \right) \bar{v} \, dS = 0.$$

Since v is arbitrary, u must satisfy (3.6) and (3.7). Finally, we use (3.18) and (3.17) in (3.9). Since $V \in \mathbb{C}^L$ is arbitrary, we must have that

$$(3.19) \quad - \int_{e_l} u \, dS + |e_l| U_l = z_l I_l, \quad 1 \leq l \leq L.$$

This, in turn, can be used together with (3.6) to obtain that

$$(3.20) \quad \int_{e_l} \sigma \frac{\partial u}{\partial \nu} \, dS = \frac{1}{z_l} \int_{e_l} (U_l - u) \, dS = I_l.$$

Thus u also satisfies (3.8). $\square$

We obtain our existence and uniqueness result from the Lax–Milgram lemma, which reads as follows [5].

LEMMA (Lax–Milgram). *Let X be a Hilbert space. Let B be a complex-valued functional defined on the product space $X \times X$ satisfying the following conditions:*

(a) *sesquilinearity:*

$$B(\alpha_1 a_1 + \alpha_2 a_2, b) = \alpha_1 B(a_1, b) + \alpha_2 B(a_2, b),$$

$$B(a, \beta_1 b_1 + \beta_2 b_2) = \bar{\beta}_1 B(a, b_1) + \bar{\beta}_2 B(a, b_2);$$

(b) *boundedness:* *There exists a positive constant γ such that $|B(a, b)| \leq \gamma \|a\| \|b\|$;*

(c) *coercivity:* *There exists a positive constant δ such that $|B(a, a)| \geq \delta \|a\|^2$.*

Then, for any continuous linear functional $f: X \rightarrow \mathbb{C}$, there is a unique b in X such that $f(a) = B(a, b)$.

Unfortunately, our bilinear form (3.10) does not satisfy hypothesis (c) of the Lax–Milgram lemma. We see the difficulty by computing

$$(3.21) \quad B((u, U), (u, U)) = \int_{\Omega} \sigma |\nabla u|^2 dx + \sum_{l=1}^L \frac{1}{z_l} \int_{e_l} |u - U_l|^2 dS.$$

Setting $B((u, U), (u, U)) = 0$ does not imply that $(u, U) = 0$, but rather that

$$(3.22) \quad u = \text{const} = U_1 = \cdots = U_L.$$

This suggests that we introduce the quotient space

$$(3.23) \quad \dot{H} = H/\mathbb{C};$$

i.e., the elements $(u, U) \in H$ and $(v, V) \in H$ are equivalent if

$$(3.24) \quad u - v = \text{const} = U_1 - V_1 = \cdots = U_L - V_L.$$

The space $\dot{H}$ is equipped in the usual way with the quotient norm

$$(3.25) \quad \|(u, U)\| = \inf_{c \in \mathbb{C}} (\|u - c\|_{H^1(\Omega)}^2 + \|U - c\|_{\mathbb{C}^L}^2)^{1/2}.$$

Showing that B of Proposition 3.1 satisfies the hypotheses of the Lax–Milgram lemma, however, is much easier with a different norm in $\dot{H}$, namely,

$$(3.26) \quad \|(u, U)\|_* = \left(\|\nabla u\|_{L^2(\Omega)}^2 + \sum_{l=1}^L \int_{e_l} |u(x) - U_l|^2 dS \right)^{1/2}.$$

The following result shows that we can use either norm.

LEMMA 3.2. *The norms (3.25) and (3.26) are equivalent; i.e., for some constants $0 < \lambda < \Lambda < \infty$,*

$$(3.27) \quad \lambda \|(u, U)\|_* \leq \|(u, U)\| \leq \Lambda \|(u, U)\|_*$$

for all $(u, U) \in \dot{H}$.

Proof. Let us first prove the first inequality. For $(u, U) \in \dot{H}$, we choose a constant $c \in \mathbb{C}$ such that

$$(3.28) \quad \|u - c\|_{H^1(\Omega)}^2 + \|U - c\|_{\mathbb{C}^L}^2 < \|(u, U)\|^2 + \varepsilon,$$

$\varepsilon > 0$ being arbitrary. Now we modify the right side of (3.26) to obtain that

$$(3.29) \quad \begin{aligned} \|(u, U)\|_*^2 &= \|\nabla(u - c)\|_{L^2(\Omega)}^2 + \sum_{l=1}^L \int_{e_l} |u(x) - c - (U_l - c)|^2 dS \\ &\leq \|\nabla(u - c)\|_{L^2(\Omega)}^2 + 2 \sum_{l=1}^L \int_{e_l} |u(x) - c|^2 dS + 2 \sum_{l=1}^L |e_l| |U_l - c|^2, \end{aligned}$$

where we have used $2ab \leq a^2 + b^2$. By the continuous imbedding $H^{1/2}(\partial\Omega) \subset L^2(\partial\Omega)$ and the trace theorem [4], the second term on the right side of (3.29) is bounded by

$$(3.30) \quad \begin{aligned} 2 \sum_{l=1}^L \int_{e_l} |u(x) - c|^2 dS &\leq 2 \|u - c\|_{L^2(\partial\Omega)}^2 \leq C \|u - c\|_{H^{1/2}(\partial\Omega)}^2 \\ &\leq C \|u - c\|_{H^1(\Omega)}^2. \end{aligned}$$

Using this to combine the first and second terms of (3.29), we obtain that

$$\|(u, U)\|_*^2 \leq C (\|u - c\|_{H^1(\Omega)}^2 + \|U - c\|_{\mathbb{C}^L}^2).$$

By (3.28), this is

$$(3.31) \quad \|(u, U)\|_*^2 \leq C (\|(u, U)\|^2 + \varepsilon).$$

Since this holds for arbitrary $\varepsilon > 0$, the first part of inequality (3.27) is proved.

To prove the second part of inequality (3.27), assume that the claim is not true. Then we can pick a sequence (u^n, U^n) , $n = 1, 2$ of elements in $\dot{H}$, so that

$$(3.32) \qquad \qquad \qquad \|(u^n, U^n)\| = 1, \qquad \|(u^n, U^n)\|_* < \frac{1}{n}.$$

Let $(c^n)_{n=1}^\infty$ be a sequence of constants chosen so that the sequence $(w^n, W^n) = (u^n - c^n, U^n - c^n)$ satisfies

$$(3.33) \qquad \qquad \qquad 1 \leq \|w^n\|_{H^1(\Omega)}^2 + \|W^n\|_{\mathbb{C}^L}^2 < 1 + \frac{1}{n}.$$

Then, by the compact imbedding theorem of Sobolev spaces over bounded domains [4], we may assume that (w^n) contains a subsequence, also denoted (w^n) , with

$$(3.34) \qquad \qquad \qquad w^n \rightarrow w \qquad \text{in } L^2(\Omega) \quad \text{as } n \rightarrow \infty$$

for some $w \in L^2(\Omega)$, However, since

$$(3.35) \qquad \qquad \qquad \|\nabla w^n\|_{L^2(\Omega)} \leq \|(u^n, U^n)\|_* < \frac{1}{n},$$

(w^n) forms a Cauchy sequence in $H^1(\Omega)$, thus converging, and the limit satisfies $\nabla w = 0$, i.e., $w = \text{constant} = c_0$.

Now, from (3.35), we have that, for each $l = 1, 2, \dots, L$,

$$\begin{aligned} \frac{1}{n} &> \int_{e_l} |u^n(x) - U_l^n|^2 \, dS = \int_{e_l} |w^n(x) - c_0 + c_0 - W_l^n|^2 \, dS \\ (3.36) \qquad &= \int_{e_l} |w^n(x) - c_0|^2 \, dx - 2(\overline{W_l^n} - \overline{c_0}) \int_{e_l} (w^n(x) - c_0) \, dx + |W_l^n - c_0|^2 |e_l| \\ &\cong -2|W_l^n - c_0| \int_{e_l} |w^n(x) - c_0| \, dS + |W_l^n - c_0|^2 |e_l| \end{aligned}$$

or

$$(3.37) \qquad \qquad \qquad |e_l| |W_l^n - c_0|^2 < \frac{1}{n} + 2|W_l^n - c_0| \int_{e_l} |w^n(x) - c_0| \, dS.$$

The factor multiplying this last integral we estimate with (3.33), giving

$$\begin{aligned} |e_l| |W_l^n - c_0|^2 &\leq \frac{1}{n} + 2 \left(1 + \frac{1}{n} + |c_0| \right) |e_l|^{1/2} \|w^n - c_0\|_{L^2(\partial\Omega)} \\ &\leq \frac{1}{n} + 2C \left(1 + \frac{1}{n} + |c_0| \right) |e_l|^{1/2} \|w^n - c_0\|_{H^1(\Omega)}. \end{aligned}$$

Since w^n converges to c_0 in $H^1(\Omega)$, this shows that W_l^n converges to c_0 , as $n \rightarrow \infty$. This is a contradiction, however, since according to (3.32) and (3.25),

$$(3.38) \qquad \qquad \qquad 1 = \|(u^n, U^n)\|^2 \leq \|w^n - c_0\|_{H^1(\Omega)}^2 + \|W^n - c_0\|_{\mathbb{C}^L}^2 \rightarrow 0. \qquad \square$$

Note that the norm $\|(u, U)\|_*$ also defines an inner product in $\dot{H}$ in a natural way.

We can now apply the Lax–Milgram lemma to obtain the existence and uniqueness of solutions to the complete model.

THEOREM 3.3. Suppose that there are strictly positive constants σ_0 , σ_1 , and Z such that

$$(3.39) \quad |\sigma| \leq \sigma_1,$$

$$(3.40) \quad \operatorname{Re} \sigma \geq \sigma_0,$$

and

$$(3.41) \quad \operatorname{Re} z_l > Z \quad \text{for } l = 1, 2, \dots, L.$$

Then, for a given current pattern $(I_l)_{l=1}^L$ satisfying (2.15), there is a unique (u, U) in $\dot{H}$ satisfying

$$(3.42) \quad B((u, U), (v, V)) = \sum_{l=1}^L I_l \bar{V}_l$$

for all $(v, V) \in \dot{H}$.

Proof. We apply the Lax–Milgram lemma with $X = \dot{H}$. That conditions (b) and (c) hold follows from (3.39)–(3.41) and Lemma 3.2. We need only check that the linear mapping

$$(3.43) \quad f: (v, V) \mapsto \sum I_l \bar{V}_l$$

is well defined and continuous. If $(v, V) \sim (\tilde{v}, \tilde{V})$, then by (2.15)

$$(3.44) \quad f(v, V) = \sum I_l \bar{V}_l = \sum I_l (\bar{\tilde{V}}_l - \text{const}) = \sum I_l \bar{\tilde{V}}_l = f(\tilde{v}, \tilde{V}).$$

Also, if $c \in \mathbb{C}$ is a constant with

$$(3.45) \quad (\|v - c\|_{H^1(\Omega)}^2 + \|V - c\|_{\mathbb{C}^L}^2)^{1/2} \leq \|(v, V)\| + \varepsilon,$$

then

$$(3.46) \quad \begin{aligned} |f(v, V)| &= |\sum I_l (\bar{V}_l - c)| \leq \|I\|_{\mathbb{C}^L} \|V_l - c\|_{\mathbb{C}^L} \\ &\leq \|I\|_{\mathbb{C}^L} (\|(v, V)\| + \varepsilon), \end{aligned}$$

which gives the continuity. $\square$

COROLLARY 3.4. Suppose that the hypotheses of Theorem 3.3 are satisfied. Then, if

$$(3.47) \quad \sum_{l=1}^L U_l = 0,$$

the complete model has a unique solution; i.e., condition (3.47) picks out a unique solution.

Remark. Another way to prove uniqueness for both the complete model and shunt model is the following. We assume that there are two solutions (u, U) and $(\tilde{u}, \tilde{U})$ of (3.5)–(3.8) satisfying (2.15) and (3.47). We let

$$(3.48) \quad (v, V) = (u, U) - (\tilde{u}, \tilde{U});$$

then (v, V) satisfies (3.5)–(3.7), and

$$(3.49) \quad \int_{e_l} \sigma \frac{\partial v}{\partial \nu} dS = 0.$$

We use (3.5) and (3.6) in Green's formula

$$(3.50) \quad 0 = \int_{\Omega} \bar{v} \nabla \cdot \sigma \nabla v \, dx = \int_{\partial\Omega} \bar{v} \sigma \frac{\partial v}{\partial \nu} dS - \int_{\Omega} \sigma |\nabla v|^2 \, dx,$$

obtaining that

$$(3.51) \quad \int_{\Omega} \sigma |\nabla v|^2 dx = \sum_{l=1}^L \int_{e_l} \left(\bar{V}_l - \bar{z}_l \bar{\sigma} \frac{\partial \bar{v}}{\partial \nu} \right) \sigma \frac{\partial v}{\partial \nu} dS.$$

Using (3.49) on the right side of (3.51), we have that

$$(3.52) \quad \int_{\Omega} \sigma |\nabla v|^2 dx = - \sum_{l=1}^L \bar{z}_l \left| \sigma \frac{\partial v}{\partial \nu} \right|^2.$$

Provided that $\operatorname{Re} z_l \geq 0$ for all l , (3.52) can be satisfied only if both sides are zero. This implies that v is constant. This implies, from (3.6), that all the V 's are also equal to this same constant; (3.47) shows that this constant must be zero.

We close this section with some remarks about the proof of Theorem 3.3. First, the existence and uniqueness result can be generalized to more complicated domains. Instead of a smooth boundary, we could assume that $\partial\Omega$ is piecewise $C^{1,1}$ (see [3]). We do not give the proof of this generalization here. Second, note that the bilinear form B is strictly coercive; i.e., for $(u, U) \in \dot{H}$, we have that

$$|B((u, U), (u, U))| \geq C \|(u, U)\|^2.$$

This suggests that a finite element algorithm for numerical solution of the problem is stable. Appendix 3 shows how B can be used for numerical calculation in a simple case.

Theorem 3.3 does not apply to the shunt model because the functional B is undefined if any z_l is zero.

The existence of a unique solution to the shunt model follows similarly by considering its weak formulation

$$B_*((u, U), (v, V)) \equiv \int_{\Omega} \sigma \nabla u \cdot \nabla \bar{v} dx = \sum_{l=1}^L I_l \bar{V}_l$$

in a subspace of (3.4) in which $u = \text{const}$ on electrode e_l .

We point out that when the surface impedances z_l are zero, the current density may be no smoother than $H^{-1/2}(\partial\Omega)$, although we do not give the details here.

4. Symmetry of the complete model. In this section we use R to denote the map from currents I to voltages V given by the complete model. We show that R satisfies the expected symmetry property.

PROPOSITION 4.1. *Suppose that the hypotheses of the theorem are satisfied, so that the map R exists. Then $R^T = R$.*

Proof. Because the adjoint R^* is equal to $\bar{R}^T$, we need only show that $R^* = \bar{R}$. To do this, we show that for arbitrary X and Y in $\mathbb{C}^L$ satisfying

$$(4.1) \quad \sum_{l=1}^L X_l = \sum_{l=1}^L Y_l = 0,$$

we have that

$$(4.2) \quad \langle RX, Y \rangle = \langle X, \bar{R}Y \rangle.$$

We denote by (u, U) the solution to the complete model satisfying

$$(4.3) \quad \nabla \cdot \sigma \nabla u = 0,$$

$$(4.4) \quad \int_{e_l} \sigma \frac{\partial u}{\partial \nu} dS = X_l,$$

$$(4.5) \quad u + z_l \sigma \frac{\partial u}{\partial \nu} = U_l,$$

and by (v, V) the solution to

$$(4.6) \quad \nabla \cdot \sigma \nabla v = 0,$$

$$(4.7) \quad \int_{e_l} \sigma \frac{\partial v}{\partial \nu} dS = \bar{Y}_l,$$

$$(4.8) \quad v + z_l \sigma \frac{\partial v}{\partial \nu} = V_l.$$

We next compute the following expression in two different ways using Green's theorem. On one hand, we have that

$$(4.9) \quad \int_{\Omega} \sigma \nabla u \cdot \nabla v dx = \int_{\partial \Omega} \sigma u \frac{\partial v}{\partial \nu} dS,$$

which by (4.5) and (4.7) is

$$(4.10) \quad \int_{\Omega} \sigma \nabla u \cdot \nabla v dx = \sum_{l=1}^L U_l \bar{Y}_l - \sum_{l=1}^L \int_{e_l} z_l \sigma \frac{\partial u}{\partial \nu} \sigma \frac{\partial v}{\partial \nu} dS.$$

On the other hand, we have that

$$(4.11) \quad \int_{\Omega} \sigma \nabla u \cdot \nabla v dx = \int_{\partial \Omega} v \sigma \frac{\partial u}{\partial \nu} dS,$$

which by (4.8) and (4.4) is

$$(4.12) \quad \int_{\Omega} \sigma \nabla u \cdot \nabla v dx = \sum_{l=1}^L V_l X_l - \sum_{l=1}^L \int_{e_l} z_l \sigma \frac{\partial v}{\partial \nu} \sigma \frac{\partial u}{\partial \nu} dS.$$

Comparison of (4.10) and (4.12) yields (4.2). $\square$

Appendix 1. This appendix shows that if a two-dimensional experiment is invariant under discrete rotations, the characteristic current patterns are the trigonometric functions (1.3).

PROPOSITION A1.1. *The discrete rotation operator Q , which acts by*

$$(A1.1) \quad Q(I_1, \dots, I_L)^T = (I_L, I_1, \dots, I_{L-1})^T,$$

has eigenvalues

$$(A1.2) \quad \lambda_L = \exp(i\theta_l), \quad \theta_l = 2l\pi/L, \quad l = 1, 2, \dots, L$$

and corresponding eigenvectors

$$(A1.3) \quad a^l = (\exp[i(L-1)\theta_l], \exp[i(L-2)\theta_l], \dots, \exp[i\theta_l], 1).$$

Proof. Observe that the a^l are independent for $l = 1, 2, \dots, L$, and that $Qa^l = \lambda_l a^l$.

PROPOSITION A1.2. *A two-dimensional experiment invariant under discrete rotations and with an even number L of electrodes has a resistance matrix R with one nonzero, nondegenerate eigenvalue and $L/2 - 1$ eigenvalues having a twofold degeneracy. The corresponding eigenvectors are*

$$(A1.4) \quad c^l = (\cos \theta_l, \cos 2\theta_l, \dots, \cos (L-1)\theta_l, 1) \quad \text{for } l = 1, 2, \dots, L/2,$$

$$(A1.5) \quad s^l = (\sin \theta_l, \sin 2\theta_l, \dots, \sin (L-1)\theta_l, 0) \quad \text{for } l = 1, 2, \dots, L/2 - 1.$$

Proof. The experiment being invariant under discrete rotations means that if we apply a rotated current pattern and measure the corresponding voltages, the result is a rotated voltage pattern, as follows:

$$(A1.6) \quad RQI = QRI.$$

Because this is true for every current pattern, we have that

$$(A1.7) \quad RQ = QR.$$

We find the eigenvectors of R by applying (A1.7) to an eigenvector a^l of Q , as follows:

$$(A1.8) \quad QRa^l = RQa^l = \lambda_l Ra^l.$$

This implies that Ra_l is also an eigenvector of Q corresponding to the eigenvalue λ_l . Because the eigenvalues of Q are nondegenerate, Ra^l must be a multiple of a^l , say,

$$(A1.9) \quad Ra^l = \alpha_l a^l.$$

We now take the complex conjugate of (A1.7). Because Q is real, we find that $\bar{R}$ also commutes with Q . The above argument then shows that, for some $\tilde{\alpha}_l$,

$$(A1.10) \quad \bar{R}a^l = \tilde{\alpha}_l a^l.$$

Taking the complex conjugate of (A1.10) gives us

$$(A1.11) \quad R\bar{a}^l = \bar{\alpha}_l \bar{a}^l.$$

Moreover, the following argument shows that $\tilde{\alpha}_l = \bar{\alpha}_l$. We take the inner product of (A1.9) with a^l , as follows:

$$(A1.12) \quad \langle Ra^l, a^l \rangle = \alpha_l |a^l|^2.$$

The left side of (A1.12), however, is equal to

$$(A1.13) \quad \langle a^l, R^* a^l \rangle,$$

which, by the symmetry of R , is equal to

$$(A1.14) \quad \langle a^l, \bar{R}a^l \rangle = \bar{\alpha}_l |a^l|^2$$

from (A1.10). This shows that $\tilde{\alpha}_l = \bar{\alpha}_l$.

Finally, adding and subtracting (A1.9) and (A1.11) gives us

$$(A1.15) \quad Rc^l = \alpha_l c^l,$$

$$(A1.16) \quad Rs^l = \alpha_l s^l,$$

where

$$(A1.17) \quad c^l = (\cos[(L-1)\theta_l], \cos[(L-2)\theta_l], \dots, \cos\theta_l, 1),$$

$$(A1.18) \quad s^l = (\sin[(L-1)\theta_l], \sin[(L-2)\theta_l], \dots, \sin\theta_l, 0).$$

Because $(L-m)\theta_l = -m\theta_l \pmod{2\pi}$ and cosine is even, we have (A1.4). The sine function is odd, but we can factor out an overall -1 from (A1.18); thus we have (A1.5).

Appendix 2. Here we carry out the second derivation of (2.9), in which we obtain a complex “conductivity.” First, we look for solutions to Maxwell’s equations of the form $B = \tilde{B} \exp(i\omega t)$, $D = \tilde{D} \exp(i\omega t)$, and so forth. Equations (2.1) and (2.2) are then

$$(A2.1) \quad \nabla \wedge E = -i\omega B,$$

$$(A2.2) \quad \nabla \wedge H = J + i\omega D,$$

where we have dropped the tildes. These equations hold even in dispersive media, where (2.5)–(2.7) are true only in the frequency domain.

Next, we expand all fields as power series in the magnetic permeability μ , as follows:

$$(A2.3) \quad E = E_0 + \mu E_1 + \cdots,$$

$$(A2.4) \quad H = H_0 + \mu H_1 + \cdots,$$

and so forth. From (A2.1) and (2.6), we have the zero-order equation

$$(A2.5) \quad \nabla \wedge E_0 = 0,$$

which implies that

$$(A2.6) \quad E_0 = -\nabla u.$$

Taking the divergence of (A2.2) and using (2.5) and (2.7) gives us the zero-order equation

$$(A2.7) \quad 0 = \nabla \cdot (\sigma + i\omega\epsilon) \nabla u.$$

Appendix 3. Numerical computation in the N -ring case. In this appendix we demonstrate the use of the coercive bilinear form (3.10) obtained in the paper to compute numerically the characteristic resistances in a simple geometry.

Assume that we have a circular body Ω with radius $r = 1$ in two dimensions, and we attach L identical, equally spaced electrodes to its boundary. We assume that L is an even number. The midpoints of the electrodes can be chosen as

$$(A3.1) \quad \theta_l = \frac{2\pi}{L} l, \quad l = 1, 2, \dots, L.$$

Let I_l denote the current applied to the l th electrode. Using the obvious relabelling of the electrodes by negative indices as $I_{-l} = I_{L-l}$, $l = 0, 1, \dots, N-1$, we may express the symmetries of the sine and cosine patterns (1.3) as

$$(A3.2) \quad I_l = I_{-l} \quad (\text{cosine}), \quad I_l = -I_{-l} \quad (\text{sine}).$$

In this appendix, we assume that the current pattern satisfies either one of these symmetries.

If the conductivity distribution σ is rotationally invariant, the surface impedances z_l are identical, i.e., $z_l = z$. It is assumed that here z is real and positive.

With these assumptions, the electric potential is either an even or an odd function of the angle θ ; i.e., we have that $u(r, -\theta) = u(r, \theta)$ (cosine pattern) or $u(r, -\theta) = -u(r, \theta)$ (sine pattern). The voltage U_l follows the same symmetry as the current pattern.

Assume that the body Ω consists of N rings with constant real-valued conductivity; i.e., we have that

$$(A3.3) \quad \sigma = \sigma(r) = \sigma_n > 0 \quad \text{for } r_{n-1} \leq r < r_n,$$

where $0 = r_0 < r_1 < \dots < r_N = 1$. Depending on whether the applied current pattern is even or odd, in each ring we may write

$$(A3.4) \quad u(r, \theta) = \sum_{m=1}^{\infty} \frac{1}{\sqrt{m}} \left(a_m^{(n)} \left(\frac{r_n}{r} \right)^m + b_m^{(n)} \left(\frac{r}{r_n} \right)^m \right) \begin{cases} \cos m\theta, \\ \sin m\theta, \end{cases}$$

where $r_{n-1} \leq r < r_n$, $1 \leq n \leq N$. The factor $1/\sqrt{m}$ is included because we know that u restricted to any circle $r = \text{const}$ is an $H^{1/2}$ function; we have written (A3.4) so that the coefficients $(a_m^{(n)})_{m=1}^{\infty}$, $(b_m^{(n)})_{m=1}^{\infty}$ are square summable. So that u is finite at the origin, we must have that $a_m^{(1)} = 0$.

Our objective is to express u in terms of the boundary values of its Fourier coefficients at $r = 1$, and then to solve for these boundary values and for the voltages U_l by a variational method. Therefore we first match the coefficients $(a_m^{(n)}, b_m^{(n)})$ at the discontinuities of σ by requiring that

$$(A3.5) \quad [u]_{r=r_n} = \left[\sigma \frac{\partial u}{\partial r} \right]_{r=r_n} = 0, \quad 1 \leq n \leq N-1.$$

Here $[\cdot]_{r=r_n}$ denotes the jump of the function of $r = r_n$. A straightforward calculation gives the recursive formula

$$(A3.6) \quad \begin{pmatrix} a_m^{(n+1)} \\ b_m^{(n+1)} \end{pmatrix} = F_m^{(n)} \begin{pmatrix} a_m^{(n)} \\ b_m^{(n)} \end{pmatrix}, \quad 1 \leq n \leq N-1,$$

where

$$(A3.7) \quad F_m^{(n)} = \frac{1}{2} \begin{pmatrix} \left(1 + \frac{\sigma_n}{\sigma_{n+1}}\right) \left(\frac{r_n}{r_{n+1}}\right)^m & \left(1 - \frac{\sigma_n}{\sigma_{n+1}}\right) \left(\frac{r_n}{r_{n+1}}\right)^m \\ \left(1 - \frac{\sigma_n}{\sigma_{n+1}}\right) \left(\frac{r_{n+1}}{r_n}\right)^m & \left(1 + \frac{\sigma_n}{\sigma_{n+1}}\right) \left(\frac{r_{n+1}}{r_n}\right)^m \end{pmatrix}.$$

From the condition $a_m^{(1)} = 0$, it follows that

$$(A3.8) \quad \begin{pmatrix} a_m^{(n)} \\ b_m^{(n)} \end{pmatrix} = b_m^{(1)} \begin{pmatrix} \alpha_m^{(n)} \\ \beta_m^{(n)} \end{pmatrix},$$

where

$$(A3.9) \quad \begin{pmatrix} \alpha_m^{(1)} \\ \beta_m^{(1)} \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \begin{pmatrix} \alpha_m^{(n+1)} \\ \beta_m^{(n+1)} \end{pmatrix} = F_m^{(n)} \begin{pmatrix} \alpha_m^{(n)} \\ \beta_m^{(n)} \end{pmatrix}, \quad 1 \leq n \leq N-1.$$

In particular, if we denote by C_m the sum

$$(A3.10) \quad C_m = a_m^{(N)} + b_m^{(N)} = b_m^{(1)} (\alpha_m^{(N)} + \beta_m^{(N)}),$$

(A3.8) can be written as

$$(A3.11) \quad \begin{pmatrix} a_m^{(n)} \\ b_m^{(n)} \end{pmatrix} = \frac{C_m}{\alpha_m^{(N)} + \beta_m^{(N)}} \begin{pmatrix} \alpha_m^{(n)} \\ \beta_m^{(n)} \end{pmatrix}.$$

The only unknowns to be found are thus the C_m . To determine the coefficients C_m and the voltages U_l , we use the well-known variational principle (see, e.g., [7]), stating that the solution to the problem of finding (u, U) satisfying

$$(A3.12) \quad B((u, U), (v, V)) = \sum_{l=1}^L I_l V_l$$

for all $(v, V) \in \dot{H}$ is tantamount to minimizing the quadratic functional

$$(A3.13) \quad \phi(u, U) = \frac{1}{2} B((u, U), (u, U)) - \sum_{l=1}^L I_l U_l.$$

Next, we can express $\phi(u, U)$ in terms of C_m 's and U_l 's by substituting formula (A3.11) into (A3.4) and calculating the integrals appearing in B . We get that

$$(A3.14) \quad \int_D \sigma |\nabla u|^2 dx = \sum_{m=1}^{\infty} \eta_m C_m^2,$$

where

$$(A3.15) \quad \eta_m = \frac{\pi}{(\alpha_m^{(N)} + \beta_m^{(N)})^2} \sum_{n=1}^N \sigma_n \left\{ \left(\left(\frac{r_n}{r_{n-1}} \right)^{2m} - 1 \right) (\alpha_m^{(n)})^2 + \left(1 - \left(\frac{r_{n-1}}{r_n} \right)^{2m} \right) (\beta_m^{(n)})^2 \right\}.$$

Furthermore, by using the symmetries of the voltages and the potential, we get that

$$(A3.16) \quad \sum_{l=1}^L \frac{1}{z} \int_{e_l} |u(1, \theta) - U_l|^2 d\theta = \sum_{l=-L/2+1}^{L/2} \frac{1}{z} \int_{e_l} |u(1, \theta) - U_l|^2 d\theta = \sum_{l=0}^{L/2} \frac{t_l}{z} \int_{e_l} |u(1, \theta) - U_l|^2 d\theta,$$

where $t_0 = 1 = t_{L/2}$, $t_l = 2$ for $l = 2, \dots, L/2 - 1$, and, by a straightforward calculation,

$$(A3.17) \quad \int_{e_l} |u(1, \theta) - U_l|^2 d\theta = \Delta U_l^2 - 2 \sum_{m=1}^{\infty} U_l \beta_{l,m} C_m + \sum_{m,m'=1}^{\infty} C_m \gamma_{m,m'}^{(l)} C_{m'},$$

where Δ is the angular width of an electrode, where

$$(A3.18) \quad \beta_{l,m} = \frac{2}{m^{3/2}} \sin m \frac{\Delta}{2} \begin{cases} \cos m\theta_l, \\ \sin m\theta_l, \end{cases}$$

and, finally,

$$(A3.19) \quad \gamma_{m,m'}^{(l)} = \frac{1}{\sqrt{mm'}} \left(\frac{\cos(m-m')\theta_l \sin(m-m')(\Delta/2)}{m-m'} \pm \frac{\cos(m+m')\theta_l \sin(m+m')(\Delta/2)}{m+m'} \right).$$

Since the U_l 's and I_l 's have the same symmetries, we have that

$$(A3.20) \quad \sum_{l=1}^L I_l U_l = \sum_{l=-L/2+1}^{L/2} I_l U_l = \sum_{l=0}^{L/2} t_l I_l U_l,$$

with t_l defined as in (A3.16).

These calculations give rise to the following discrete minimization problem, equivalent to minimizing the functional ϕ in $\dot{H}$: Find a square summable vector $x = (x_0, x_1, \dots)$ that minimizes the expression $\Psi(x) = x^T S x - x^T r$, where

$$S = \begin{pmatrix} \frac{\Delta}{z} \text{diag}(t) & -\frac{1}{z} (\beta_{l,m})_{\substack{0 \leq l \leq L/2 \\ 1 \leq m < \infty}} \\ -\frac{1}{z} (\beta_{m,l})_{\substack{1 \leq m < \infty \\ 0 \leq l \leq L/2}} & \text{diag}(\eta) + \frac{1}{z} \left(\sum_{l=0}^{L/2} t_l \gamma_{m,m'}^{(l)} \right)_{\substack{1 \leq m < \infty \\ 1 \leq m' \leq \infty}} \end{pmatrix}$$

and

$$r = (I_0, 2I_1, \dots, 2I_{L/2-1}, I_{L/2}, 0, 0, \dots)^T.$$

The unknown variables U_l and C_m are then obtained as

$$U_l = x_l, \quad l = 0, 1, \dots, L/2, \\ C_m = x_{L/2+m}, \quad 1 \leq m < \infty.$$

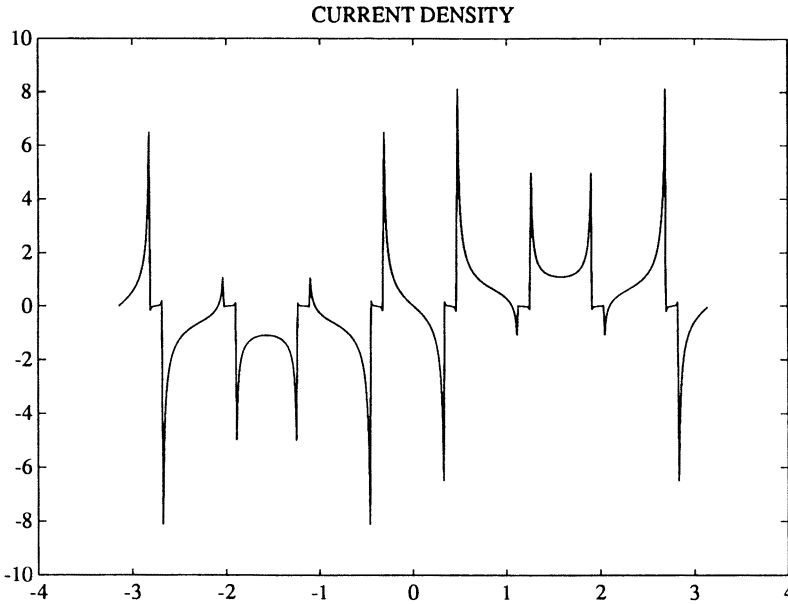


FIG. 2. This shows the current density as a function of angle for a homogeneous body with eight electrodes. The electrodes fill 82 percent of the circumference. The applied current pattern was $I_1 = \sin \theta_1$, and we truncated (A3.4) at $m = 512$.

In practice, S is truncated to a finite square matrix S_M of size $M \times M$, and r to a vector r_M of length M , and so $x_M = S_M^{-1}r_M$ is the solution of the truncated minimization problem. The truncation of S and r corresponds, of course, to truncating the Fourier series (A3.4). Note that, by the boundedness from below of the bilinear form B , it follows that $\|S\| \geq z$, and the minimization problem is well conditioned.

Figure 2 shows the current density at the boundary as obtained by the above method. The graph was obtained by summing the truncated Fourier series.

Acknowledgments. The experimental work and calculations that yielded Fig. 1 and Table 1 were done in collaboration with K. S. Cheng [1]. We also thank J. C. Newell, D. G. Gisser, and J. C. Goble, whose assistance and collaboration have made the work possible. Finally, we thank the Academy of Finland, the Emil Aaltonen Foundation, and Rensselaer Polytechnic Institute for making Erkki Somersalo's visit at Rensselaer Polytechnic Institute possible.

REFERENCES

- [1] K.-S. CHENG, D. ISAACSON, J. C. NEWELL, AND D. G. GISSER, *Electrode models for electric current computed tomography*, IEEE Trans. Biomed. Engrg. 3, 36 (1989), pp. 918–924.
- [2] D. G. GISSER, D. ISAACSON, AND J. C. NEWELL, *Current topics in impedance imaging*, Clin. Phys. Physiol. Meas., 8 (1987), pp. 39–46.
- [3] P. GRISVARD, *Elliptic Problems in Nonsmooth Domains*, Pitman, Boston, 1985.
- [4] J. L. LIONS AND E. MAGENES, *Non-Homogeneous Boundary Value Problems*, Springer, New York, 1972.
- [5] K. YOSIDA, *Functional Analysis*, 6th ed., Springer, New York, 1980.
- [6] J. OOSTENDORP AND A. VAN OOSTEROM, *The potential distribution generated by surface electrodes in inhomogeneous volume conductors of arbitrary shape*, IEEE Trans. Biomed Engrg., 38 (1991), pp. 409–417.
- [7] R. COURANT, AND D. HILBERT, *Methods of Mathematical Physics*, Interscience, New York, 1953.